Ch#9: Relations

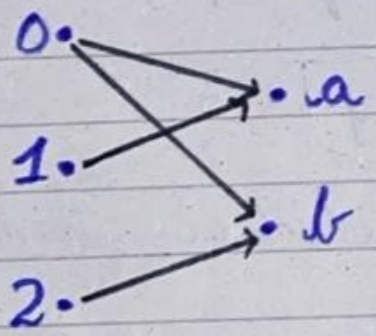
What is a relation?

- A relation R from set A to set B is a subset of the Cartesian product $A \times B$. That means.

$$R \subseteq A \times B$$

- If $(a, b) \in R$, we say "a is related to b" via relation R, written as aRb.

* Example 1: Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $\{(0,a), (0,b), (1,a), (2,b)\}$ is a relation from A to B. Relations can be represented graphically and by use of table.



R	a	b
0	×	×
1	×	
2		×

$n(A) = m$
$n(B) = n$
$n(A \times B) = m \times n$

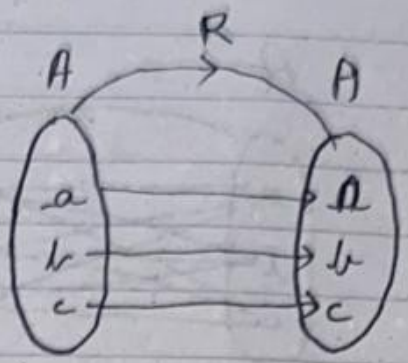
no of subsets
$(A \times B) = 2^{m \times n}$
$(A \times A) = 2^{n^2}$

Types of Relations :-

1) Identity Relation :-

A relation 'R' on set 'A' is said to be identity relation if every element of set 'A' is related to itself only.

$A = \{a, b, c\}$



$$I = \{(a,a) \mid a \in A\}$$

* Always reflexive, symmetric and transitive.

Q Example: if $A = \{1, 2, 3\}$. then

sol:

$$I = \{(1,1), (2,2), (3,3)\}$$

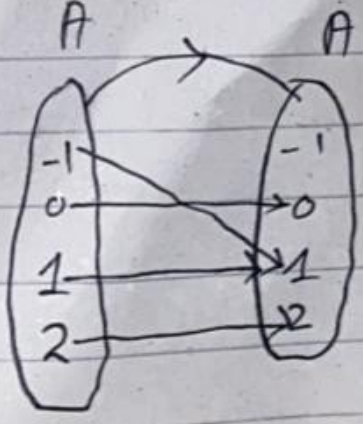
* No of Identity Relation that can be defined on set 'A' is 1.

Q: Consider a relation 'R' on set $A = \{-1, 0, 1, 2\}$ where $R = \{(a,b) : |a| = b \text{ and } a, b \in A\}$. Is it Identity ?

$$R = \{(-1,1), (0,0), (1,1), (2,2)\}$$

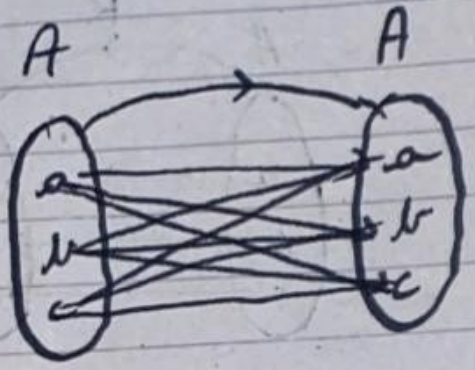
not same

not identity.



2) Universal Relation :-

$A = \{a, b, c\}$



$R = \{(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)\}$.

- A relation $R = A \times A$ is said to be universal relation on set 'A'.
- Always reflexive, symmetric and transitive.

3) Empty (Void) Relation :-

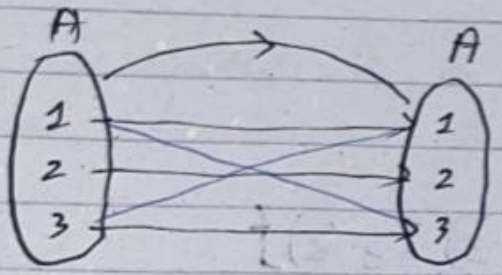
4) Reflexive & Irreflexive :-

a) Reflexive Relation :-

A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$. In other words, $\forall a ((a,a) \in R)$.

* All self pairs must be present.

$A = \{1, 2, 3\}$



→ Reflexive

Har element apne aap se related hona chahie bss ye condition hai baki or jitne relations hon.

* No of reflexive relations that can be defined from A to A = 2^{n^2-n}, if $n(A) = n$

Irreflexive Relation :-

A relation R on a set A is called irreflexive if :

$$\forall a \in A, (a, a) \notin R$$

That means no element is related to itself.

Example 1 :
Let $A = \{1, 2, 3\}$

$$R = \{(1,2), (2,3), (3,1)\}$$

Irreflexive no self pair is present.

Example 2 :

$$R = \{(1,1), (2,3)\}$$

Not irreflexive (1, 1) is a self pair.

★ If even one self pair (a, a) is found, the relation is not reflexive.

★ A relation can be neither reflexive nor irreflexive if some self-pairs are missing and some are present.

5) Symmetric & Antisymmetric

Symmetric Relation :-

Let A be a set and R a relation on A. Then, R is symmetric if:

$$\forall a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$$

Meaning : Every time you have a pair (a, b), you must also have the pair (b, a).

"If a is related to b", then "b must be related to a".

Example 1 :

Let $A = \{1, 2, 3\}$

$$R = \{(1,2), (2,1), (2,3), (3,2), (1,1)\}$$

All pairs have their reverse → Symmetric.

2) $$R = \{(1,2), (2,3), (3,1)\}$$

No reverse pairs like $(2,1), (3,2), (1,3)$
→ Not Symmetric..

Antisymmetric Relation :-

. Formal Definition :-

Let A be a set and R a relation on A.
Then R is antisymmetric if

$$\forall a, b \in A, [(a,b) \in R \text{ and } (b,a) \in R] \rightarrow a = b$$

Means:

"If both (a,b) and (b,a) are in a relation, then it must be the case that $a = b$."

So, Only self pairs like (a,a) are allowed.
if $a \neq b$, you can only have one direction.
not both.

. Example: Let $A = \{1,2,3\}$

1) $R = \{(1,1),(2,2),(3,3),(1,2)\}$

✓ $(1,2)$ is present, but not $(2,1)$
✓ All self pairs allowed
✓ So Antisymmetric.

) $R = \{(1,2),(2,1),(3,3)\}$

Not Antisymmetric.

Transitive Relation

* Definition :-

A relation R on a set A is called transitive if :

$\forall a, b, c \in A, (a, b) \in R$ and $(b, c) \in R \rightarrow (a, c) \in R$

This means if you can go from $a \rightarrow b$ and $b \rightarrow c$, then $a \rightarrow c$ must also be in the relation.

- Example 1 (Transitive) :

$$R = \{(1, 2), (2, 3), (1, 3)\}$$

Transitive

Example 2 (Not transitive) :

$$R = \{(1, 2), (2, 3)\}$$

Not transitive – (1, 3) is missing.

To test for transitivity :

- Look at every pair (a, b), (b, c)
- Check if the jump pair (a, c) is also in the relation.